Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Delta Woods Dairy

Physical address of dairy:

18700 S Bacchetti RD	Tracy	San Joaquin	95304
Number and Street	City	County	Zip Code

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1944

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X212-X110-XX08-XXXX X212-X120-XX02-XXXX X212-X120-XX03-XXXX X212-X120-XX09-XXXX X212-X120-XX14-XXXX

B. OPERATORS

Vander Woude, Arlan

Operator name: Vander Woude, Arlan

Telephone no.: Landline: ; Cellular: (209) 564-2624

18700 S Bacchetti RD	Tracy	CA	95304
Mailing Address Number and Street	City	State	Zip Code

This operator is responsible for paying permit fees.

C. OWNERS

Vander Woude, Arlan

Legal owner name: Vander Woude, Arlan

Telephone no.: Landline: ; Cellular: (209) 564-2624

18700 S Bacchetti RD	Tracy	CA	95304
Mailing Address Number and Street	City	State	Zip Code

This owner is responsible for paying permit fees.

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	220	200	110	0
Number under roof	670	90	0	0	0	0
Maximum number	700	120	220	200	120	0
Average number	690	90	210	200	110	0
Avg live weight (lbs)	1,400	1,450	1,000	750		

Predominant milk cow breed: Holstein

Average milk production: 72 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 23,680.56 *tons per reporting period*

Total nitrogen from manure: 294,484.82 *lbs per reporting period*

After ammonia losses (30% loss applied): 206,139.37 *lbs per reporting period*

Total phosphorus from manure: 48,716.25 *lbs per reporting period*

Total potassium from manure: 133,048.03 *lbs per reporting period*

Total salt from manure: 345,582.00 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 10,500,000 *gallons*

Total nitrogen generated: 54,858.69 *lbs*

Total phosphorus generated: 5,317.58 *lbs*

Total potassium generated: 89,969.78 *lbs*

Total salt generated: 473,652.19 *lbs*

	10,500,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	10,500,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
River	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

Date	Material type / Description	Quantity	Reporting basis	Moisture (%)	N (%)	P (%)	K (%)	Salt (%)
06/07/2023	Solid commercial fertilizer *4-0-0*	22.40 *ton*	Dry-weight	0.1	4.000000	0.000000	0.000000	0.000000
06/08/2023	Solid commercial fertilizer *UN32*	10.60 *ton*	Dry-weight	0.1	32.000000	0.000000	0.000000	0.000000

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	8,567.42	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total imports for all materials	8,567.42	0.00	0.00	0.00

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Field A1	15	15	2	manure	X212-X110-XX06-XXXX
Field A2	14	14	2	manure	X212-X120-XX01-XXXX
Field A3	39	39	2	manure	X212-X120-XX01-XXXX
Field A4	45	45	2	manure	X212-X120-XX01-XXXX
Field A5	5	5	0	none	X212-X120-XX01-XXXX
Field B1	15	15	1	manure	X212-X120-XX05-XXXX X212-X120-XX09-XXXX X212-X120-XX10-XXXX X212-X120-XX11-XXXX X212-X120-XX12-XXXX X212-X120-XX13-XXXX
Field B2	38	38	2	process wastewater	X212-X120-XX08-XXXX X212-X120-XX09-XXXX
Field B3	74	74	2	process wastewater	X212-X120-XX08-XXXX
Field D1	39	39	1	manure	X212-X110-XX08-XXXX
Field D2	55	55	1	manure	X212-X110-XX08-XXXX
Totals for areas that were used for application	334	334	15		
Totals for areas that were not used for application	5	5	0		
Land application area totals	339	339	15		

B. CROPS AND HARVESTS

Field A1

Field name: Field A1

Field A1

11/16/2022: Oats, hay

Crop: Oats, hay Acres planted: 15 Plant date: 11/16/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/12/2023	71.00 *ton*	As-is		16.6	24,000.00	2,160.00	19,900.00		8.86

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	4.00	160.00	26.00	132.00	0.00
Total actual harvest content	4.73	227.20	20.45	188.39	699.51

06/07/2023: Corn, silage

Crop: Corn, silage Acres planted: 15 Plant date: 06/07/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/09/2023	440.00 *ton*	As-is		72.6	3,340.00	810.00	3,340.00		6.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	26.00	208.00	39.00	171.60	0.00
Total actual harvest content	29.33	195.95	47.52	195.95	1,028.78

Field A2

Field name: Field A2

11/16/2022: Oats, hay

Crop: Oats, hay Acres planted: 14 Plant date: 11/16/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/12/2023	70.00 *ton*	As-is		8.0	17,000.00	2,750.00	25,900.00		9.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	4.00	160.00	26.00	132.00	0.00
Total actual harvest content	5.00	170.00	27.50	259.00	883.20

Field A2

06/07/2023: Corn, silage

Crop: Corn, silage Acres planted: 14 Plant date: 06/07/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/09/2023	406.00 *ton*	As-is		72.6	3,340.00	810.00	3,340.00		6.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	26.00	208.00	39.00	171.60	0.00
Total actual harvest content	29.00	193.72	46.98	193.72	1,017.09

Field A3

Field name: Field A3

11/16/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 39 Plant date: 11/16/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/29/2023	585.00 *ton*	As-is		69.4	4,490.00	1,040.00	7,050.00		12.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	15.00	134.70	31.20	211.50	1,110.78

05/26/2023: Corn, silage

Crop: Corn, silage Acres planted: 39 Plant date: 05/26/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/25/2023	1,092.00 *ton*	As-is		70.0	3,780.00	700.00	4,490.00		6.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	26.00	208.00	39.00	171.60	0.00
Total actual harvest content	28.00	211.68	39.20	251.44	1,058.40

Field A4

Field name: Field A4

11/16/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 45 Plant date: 11/16/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/29/2023	655.00 *ton*	As-is		75.6	3,980.00	880.00	5,420.00		12.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	14.56	115.86	25.62	157.78	880.79

05/25/2023: Corn, silage

Crop: Corn, silage Acres planted: 45 Plant date: 05/25/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/23/2023	1,260.00 *ton*	As-is		65.8	3,730.00	710.00	3,310.00		4.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	26.00	208.00	39.00	171.60	0.00
Total actual harvest content	28.00	208.88	39.76	185.36	919.30

Field B1

Field name: Field B1

11/10/2016: Almond, in shell

Crop: Almond, in shell Acres planted: 15 Plant date: 11/10/2016

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/06/2023	87.00 *ton*	As-is		11.2	17,700.00	1,250.00	20,800.00		7.16

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	6.00	258.00	43.80	279.00	0.00
Total actual harvest content	5.80	205.32	14.50	241.28	737.54

Field B2

Field name: Field B2

11/14/2022: Rye grass

Crop: Rye grass Acres planted: 37 Plant date: 11/14/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/28/2023	629.00 *ton*	As-is		70.6	5,270.00	1,100.00	11,900.00		13.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	255.00	37.50	246.00	0.00
Total actual harvest content	17.00	179.18	37.40	404.60	1,349.46

06/07/2023: Corn, silage

Crop: Corn, silage Acres planted: 37 Plant date: 06/07/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/09/2023	1,088.00 *ton*	As-is		76.2	3,310.00	170.00	720.00		7.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	26.00	208.00	39.00	171.60	0.00
Total actual harvest content	29.41	194.66	10.00	42.34	979.79

Field B3

Field name: Field B3

11/15/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 74 Plant date: 11/15/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/29/2023	1,288.00 *ton*	As-is		73.2	4,500.00	1,000.00	7,200.00		11.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	17.41	156.65	34.81	250.64	1,110.19

Field B3

05/24/2023: Corn, silage

Crop: Corn, silage Acres planted: 74 Plant date: 05/24/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/25/2023	2,220.00 *ton*	As-is		69.0	4,000.00	830.00	5,420.00		6.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	26.00	208.00	39.00	171.60	0.00
Total actual harvest content	30.00	240.00	49.80	325.20	1,283.40

Field D1

Field name: Field D1

11/10/2016: Almond, in shell

Crop: Almond, in shell Acres planted: 37 Plant date: 11/10/2016

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/06/2023	188.00 *ton*	As-is		11.2	17,700.00	1,250.00	20,800.00		7.16

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	6.00	258.00	43.80	279.00	0.00
Total actual harvest content	5.08	179.87	12.70	211.37	646.12

Field D2

Field name: Field D2

Field D2

10/30/2018: Almond, in shell

Crop: Almond, in shell Acres planted: 54 Plant date: 10/30/2018

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/13/2023	285.00 *ton*	As-is		11.2	17,700.00	1,250.00	20,800.00		7.16

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	6.00	258.00	43.80	279.00	0.00
Total actual harvest content	5.28	186.83	13.19	219.56	671.13

NUTRIENT BUDGET

A. LAND APPLICATIONS

Field A1 - 11/16/2022: Oats, hay

Field name: Field A1

Crop: Oats, hay — Plant date: 11/16/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/24/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Manure	Corral solids	231.99	89.42	327.32	2,286.16	90.00 *ton*
Application event totals		231.99	89.42	327.32	2,286.16	

Field A1 - 06/07/2023: Corn, silage

Field name: Field A1

Crop: Corn, silage — Plant date: 06/07/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/13/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Manure	Corral solids	246.76	69.22	221.36	3,565.30	90.00 *ton*
Application event totals		246.76	69.22	221.36	3,565.30	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	126.84	2,280,000.00 *gal*
Application event totals		0.00	0.00	0.00	126.84	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/07/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	8.00	0.00	0.00	0.00	
Application event totals		8.00	0.00	0.00	0.00	

Field A1 - 06/07/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	113.49	2,040,000.00 *gal*
Application event totals		0.00	0.00	0.00	113.49	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	106.82	1,920,000.00 *gal*
Application event totals		0.00	0.00	0.00	106.82	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	100.14	1,800,000.00 *gal*
Application event totals		0.00	0.00	0.00	100.14	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	93.46	1,680,000.00 *gal*
Application event totals		0.00	0.00	0.00	93.46	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	93.46	1,680,000.00 *gal*
Application event totals		0.00	0.00	0.00	93.46	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	80.11	1,440,000.00 *gal*
Application event totals		0.00	0.00	0.00	80.11	

Field A2 - 11/16/2022: Oats, hay

Field A2 - 11/16/2022: Oats, hay

Field name: Field A2

Crop: Oats, hay

Plant date: 11/16/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/24/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Manure	Corral solids	220.94	85.16	311.73	2,177.29	80.00 *ton*
Application event totals		220.94	85.16	311.73	2,177.29	

Field A2 - 06/07/2023: Corn, silage

Field name: Field A2

Crop: Corn, silage

Plant date: 06/07/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/13/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Manure	Corral solids	235.01	65.92	210.82	3,395.52	80.00 *ton*
Application event totals		235.01	65.92	210.82	3,395.52	

05/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	171.67	2,880,000.00 *gal*
Application event totals		0.00	0.00	0.00	171.67	

06/07/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	8.00	0.00	0.00	0.00	
Application event totals		8.00	0.00	0.00	0.00	

Field A2 - 06/07/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	150.21	2,520,000.00 *gal*
Application event totals		0.00	0.00	0.00	150.21	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	157.36	2,640,000.00 *gal*
Application event totals		0.00	0.00	0.00	157.36	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	143.06	2,400,000.00 *gal*
Application event totals		0.00	0.00	0.00	143.06	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	143.06	2,400,000.00 *gal*
Application event totals		0.00	0.00	0.00	143.06	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	135.90	2,280,000.00 *gal*
Application event totals		0.00	0.00	0.00	135.90	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	128.75	2,160,000.00 *gal*
Application event totals		0.00	0.00	0.00	128.75	

Field A3 - 11/16/2022: Oats, silage-soft dough

Field A3 - 11/16/2022: Oats, silage-soft dough

Field name: Field A3

Crop: Oats, silage-soft dough

Plant date: 11/16/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/24/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Separated Solids	Separator solids	149.02	39.32	54.98	2,993.02	330.00 *ton*
Application event totals		149.02	39.32	54.98	2,993.02	

Field A3 - 05/26/2023: Corn, silage

Field name: Field A3

Crop: Corn, silage

Plant date: 05/26/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/02/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Manure	Corral solids	263.63	73.95	236.49	3,809.08	250.00 *ton*
Application event totals		263.63	73.95	236.49	3,809.08	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	133.52	6,240,000.00 *gal*
Application event totals		0.00	0.00	0.00	133.52	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/26/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	8.00	0.00	0.00	0.00	
Application event totals		8.00	0.00	0.00	0.00	

Field A3 - 05/26/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	128.38	6,000,000.00 *gal*
Application event totals		0.00	0.00	0.00	128.38	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	125.82	5,880,000.00 *gal*
Application event totals		0.00	0.00	0.00	125.82	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	123.25	5,760,000.00 *gal*
Application event totals		0.00	0.00	0.00	123.25	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	118.11	5,520,000.00 *gal*
Application event totals		0.00	0.00	0.00	118.11	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	123.25	5,760,000.00 *gal*
Application event totals		0.00	0.00	0.00	123.25	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	120.68	5,640,000.00 *gal*
Application event totals		0.00	0.00	0.00	120.68	

Field A4 - 11/16/2022: Oats, silage-soft dough

Field A4 - 11/16/2022: Oats, silage-soft dough

Field name: Field A4

Crop: Oats, silage-soft dough

Plant date: 11/16/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/24/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Settling Basin	Separator solids	148.72	39.24	54.87	2,986.97	380.00 *ton*
Application event totals		148.72	39.24	54.87	2,986.97	

Field A4 - 05/25/2023: Corn, silage

Field name: Field A4

Crop: Corn, silage

Plant date: 05/25/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/02/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Manure	Corral solids	255.90	71.78	229.56	3,697.34	280.00 *ton*
Application event totals		255.90	71.78	229.56	3,697.34	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	124.62	6,720,000.00 *gal*
Application event totals		0.00	0.00	0.00	124.62	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/25/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	8.00	0.00	0.00	0.00	
Application event totals		8.00	0.00	0.00	0.00	

Field A4 - 05/25/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	120.17	6,480,000.00 *gal*
Application event totals		0.00	0.00	0.00	120.17	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	115.72	6,240,000.00 *gal*
Application event totals		0.00	0.00	0.00	115.72	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	111.27	6,000,000.00 *gal*
Application event totals		0.00	0.00	0.00	111.27	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	111.27	6,000,000.00 *gal*
Application event totals		0.00	0.00	0.00	111.27	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	106.82	5,760,000.00 *gal*
Application event totals		0.00	0.00	0.00	106.82	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	106.82	5,760,000.00 *gal*
Application event totals		0.00	0.00	0.00	106.82	

Field B1 - 11/10/2016: Almond, in shell

Field B1 - 11/10/2016: Almond, in shell

Field name: Field B1

Crop: Almond, in shell

Plant date: 11/10/2016

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/01/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	110.47	1,985,623.00 *gal*
Application event totals		0.00	0.00	0.00	110.47	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/19/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Manure	Corral solids	154.66	59.61	218.21	1,524.10	60.00 *ton*
Application event totals		154.66	59.61	218.21	1,524.10	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	220.93	3,971,246.00 *gal*
Application event totals		0.00	0.00	0.00	220.93	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/18/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	32.00	0.00	0.00	0.00	
Application event totals		32.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	220.93	3,971,246.00 *gal*
Application event totals		0.00	0.00	0.00	220.93	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	220.93	3,971,246.00 *gal*
Application event totals		0.00	0.00	0.00	220.93	

Field B1 - 11/10/2016: Almond, in shell

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/07/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	32.00	0.00	0.00	0.00	
Application event totals		32.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	220.93	3,971,246.00 *gal*
Application event totals		0.00	0.00	0.00	220.93	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	110.47	1,985,623.00 *gal*
Application event totals		0.00	0.00	0.00	110.47	

Field B2 - 11/14/2022: Rye grass

Field name: Field B2

Crop: Rye grass

Plant date: 11/14/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	239.90	46.34	289.70	2,286.44	1,920,000.00 *gal*
Application event totals		239.90	46.34	289.70	2,286.44	

Field B2 - 06/07/2023: Corn, silage

Field name: Field B2

Crop: Corn, silage

Plant date: 06/07/2023

Field B2 - 06/07/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	243.58	18.94	113.67	1,254.46	900,000.00 *gal*
River	Surface water	0.00	0.00	0.00	146.15	6,480,000.00 *gal*
Application event totals		243.58	18.94	113.67	1,400.61	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/07/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	8.00	0.00	0.00	0.00	
Application event totals		8.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	140.74	6,240,000.00 *gal*
Application event totals		0.00	0.00	0.00	140.74	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	140.74	6,240,000.00 *gal*
Application event totals		0.00	0.00	0.00	140.74	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	132.62	5,880,000.00 *gal*
Application event totals		0.00	0.00	0.00	132.62	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	135.32	6,000,000.00 *gal*
Application event totals		0.00	0.00	0.00	135.32	

Field B2 - 06/07/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	129.91	5,760,000.00 *gal*
Application event totals		0.00	0.00	0.00	129.91	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	129.91	5,760,000.00 *gal*
Application event totals		0.00	0.00	0.00	129.91	

Field B3 - 11/15/2022: Oats, silage-soft dough

Field name: Field B3

Crop: Oats, silage-soft dough Plant date: 11/15/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/16/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	207.29	16.50	877.71	3,124.91	5,520,000.00 *gal*
Application event totals		207.29	16.50	877.71	3,124.91	

Field B3 - 05/24/2023: Corn, silage

Field name: Field B3

Crop: Corn, silage Plant date: 05/24/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Field B3 - 05/24/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	292.30	22.73	136.41	1,505.35	2,160,000.00 *gal*
River	Surface water	0.00	0.00	0.00	129.91	11,520,000.00 *gal*
Application event totals		292.30	22.73	136.41	1,635.26	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/24/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	8.00	0.00	0.00	0.00	
Application event totals		8.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	124.50	11,040,000.00 *gal*
Application event totals		0.00	0.00	0.00	124.50	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	121.79	10,800,000.00 *gal*
Application event totals		0.00	0.00	0.00	121.79	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	120.44	10,680,000.00 *gal*
Application event totals		0.00	0.00	0.00	120.44	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	119.09	10,560,000.00 *gal*
Application event totals		0.00	0.00	0.00	119.09	

Field B3 - 05/24/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	116.38	10,320,000.00 *gal*
Application event totals		0.00	0.00	0.00	116.38	

09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	112.32	9,960,000.00 *gal*
Application event totals		0.00	0.00	0.00	112.32	

Field D1 - 11/10/2016: Almond, in shell

Field name: Field D1

Crop: Almond, in shell

Plant date: 11/10/2016

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/01/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	88.56	3,926,626.00 *gal*
Application event totals		0.00	0.00	0.00	88.56	

11/19/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Manure	Corral solids	156.75	60.42	221.16	1,544.70	150.00 *ton*
Application event totals		156.75	60.42	221.16	1,544.70	

04/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	177.12	7,853,252.00 *gal*
Application event totals		0.00	0.00	0.00	177.12	

Field D1 - 11/10/2016: Almond, in shell

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/19/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	32.00	0.00	0.00	0.00	
Application event totals		32.00	0.00	0.00	0.00	

05/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	177.12	7,853,252.00 *gal*
Application event totals		0.00	0.00	0.00	177.12	

06/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	177.12	7,853,252.00 *gal*
Application event totals		0.00	0.00	0.00	177.12	

06/08/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	32.00	0.00	0.00	0.00	
Application event totals		32.00	0.00	0.00	0.00	

07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	177.12	7,853,252.00 *gal*
Application event totals		0.00	0.00	0.00	177.12	

08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	88.56	3,926,626.00 *gal*
Application event totals		0.00	0.00	0.00	88.56	

Field D2 - 10/30/2018: Almond, in shell

Field D2 - 10/30/2018: Almond, in shell

Field name: Field D2

Crop: Almond, in shell

Plant date: 10/30/2018

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/01/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	97.71	6,322,966.00 *gal*
Application event totals		0.00	0.00	0.00	97.71	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/19/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Manure	Corral solids	157.52	60.72	222.25	1,552.33	220.00 *ton*
Application event totals		157.52	60.72	222.25	1,552.33	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	195.43	12,645,932.00 *gal*
Application event totals		0.00	0.00	0.00	195.43	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/19/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	32.00	0.00	0.00	0.00	
Application event totals		32.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	195.43	12,645,932.00 *gal*
Application event totals		0.00	0.00	0.00	195.43	

Field D2 - 10/30/2018: Almond, in shell

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	195.43	12,645,932.00 *gal*
River	Surface water	0.00	0.00	0.00	195.43	12,645,932.00 *gal*
Application event totals		0.00	0.00	0.00	390.85	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/08/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	32.00	0.00	0.00	0.00	
Application event totals		32.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	195.43	12,645,932.00 *gal*
Application event totals		0.00	0.00	0.00	195.43	

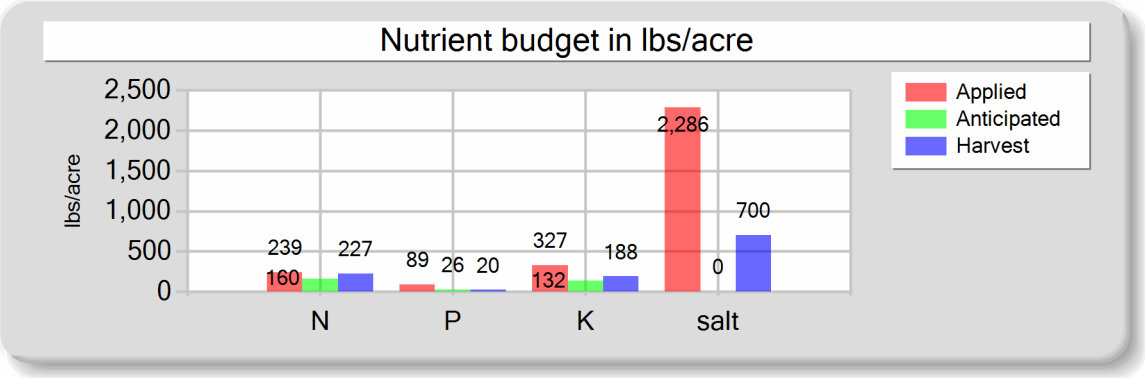
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	97.71	6,322,966.00 *gal*
Application event totals		0.00	0.00	0.00	97.71	

B. NUTRIENT BUDGET

Field A1 - 11/16/2022: Oats, hay

Field name: Field A1 Crop: Oats, hay Plant date: 11/16/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	231.99	89.42	327.32	2,286.16
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	238.99	89.42	327.32	2,286.16
Anticipated crop nutrient removal	160.00	26.00	132.00	0.00
Actual crop nutrient removal	227.20	20.45	188.39	699.51
Nutrient balance	11.79	68.97	138.93	1,586.64
Applied to removed ratio	1.05	4.37	1.74	3.27

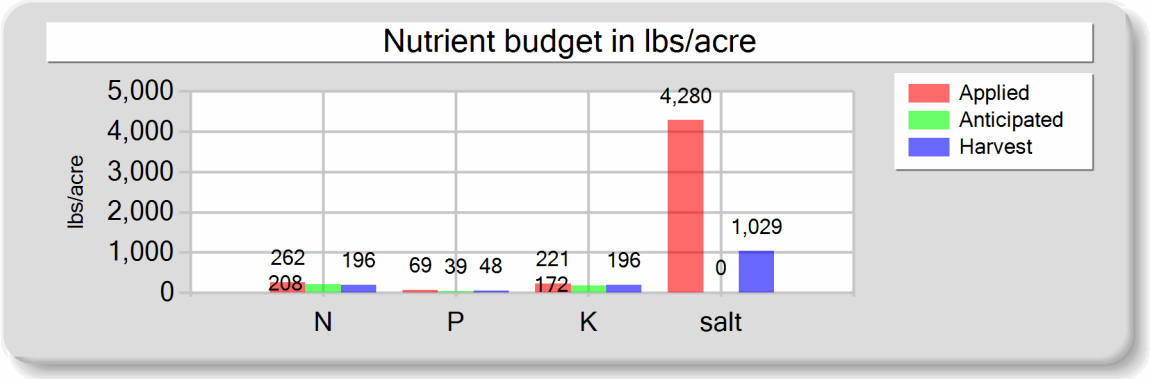
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

Field A1 - 06/07/2023: Corn, silage

Field name: Field A1 Crop: Corn, silage Plant date: 06/07/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	8.00	0.00	0.00	0.00
Dry manure	246.76	69.22	221.36	3,565.30
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	714.33
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	261.76	69.22	221.36	4,279.63
Anticipated crop nutrient removal	208.00	39.00	171.60	0.00
Actual crop nutrient removal	195.95	47.52	195.95	1,028.78
Nutrient balance	65.81	21.70	25.41	3,250.85
Applied to removed ratio	1.34	1.46	1.13	4.16

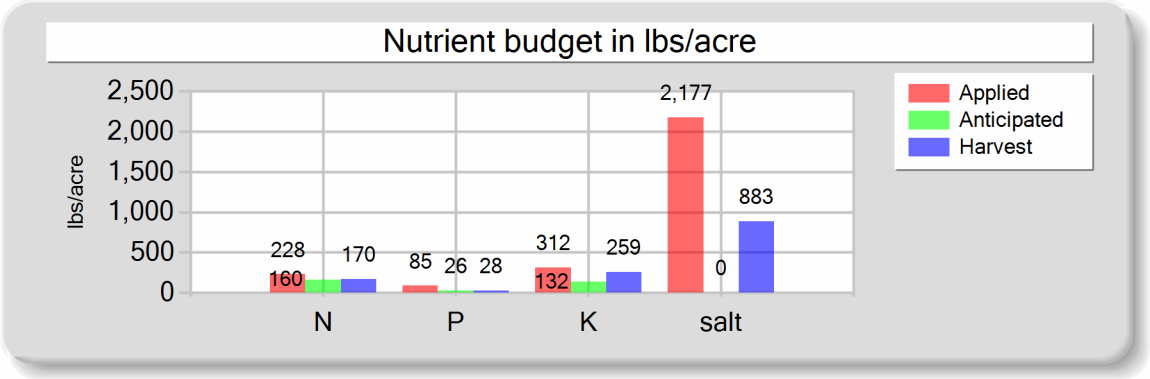
Fresh water applied	
12,840,000.00	*gallons*
472.85	*acre-inches*
31.52	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field A2 - 11/16/2022: Oats, hay

Field name: Field A2 Crop: Oats, hay Plant date: 11/16/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	220.94	85.16	311.73	2,177.29
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	227.94	85.16	311.73	2,177.29
Anticipated crop nutrient removal	160.00	26.00	132.00	0.00
Actual crop nutrient removal	170.00	27.50	259.00	883.20
Nutrient balance	57.94	57.66	52.73	1,294.09
Applied to removed ratio	1.34	3.10	1.20	2.47

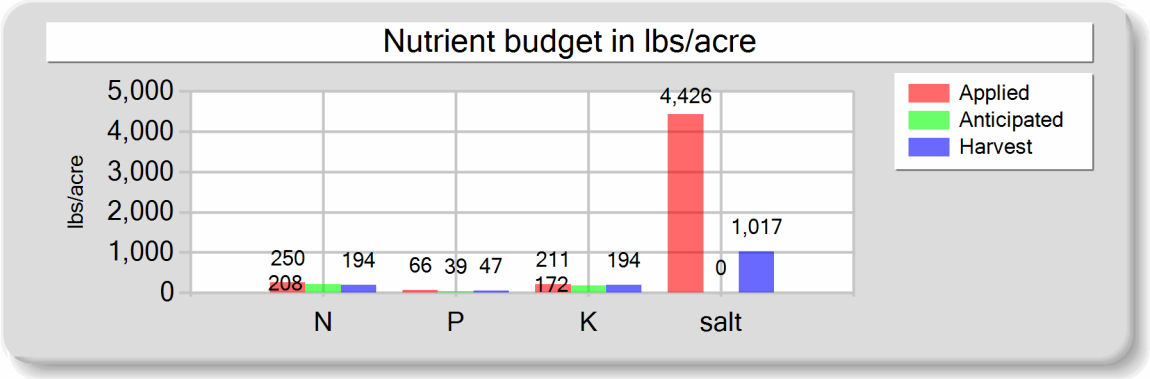
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

Field A2 - 06/07/2023: Corn, silage

Field name: Field A2 Crop: Corn, silage Plant date: 06/07/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	8.00	0.00	0.00	0.00
Dry manure	235.01	65.92	210.82	3,395.52
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	1,030.01
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	250.01	65.92	210.82	4,425.53
Anticipated crop nutrient removal	208.00	39.00	171.60	0.00
Actual crop nutrient removal	193.72	46.98	193.72	1,017.09
Nutrient balance	56.29	18.94	17.10	3,408.44
Applied to removed ratio	1.29	1.40	1.09	4.35

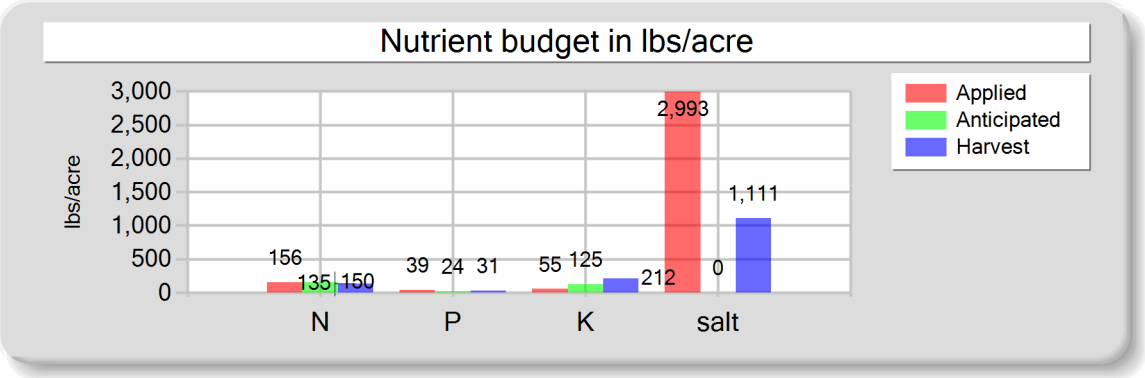
Fresh water applied	
17,280,000.00	*gallons*
636.36	*acre-inches*
45.45	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field A3 - 11/16/2022: Oats, silage-soft dough

Field name: Field A3 Crop: Oats, silage-soft dough Plant date: 11/16/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	149.02	39.32	54.98	2,993.02
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	156.02	39.32	54.98	2,993.02
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	134.70	31.20	211.50	1,110.78
Nutrient balance	21.32	8.12	-156.52	1,882.24
Applied to removed ratio	1.16	1.26	0.26	2.69

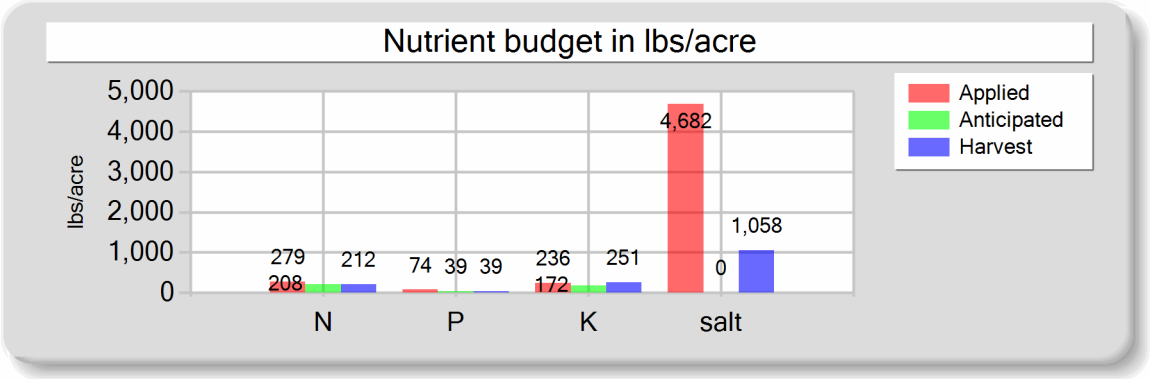
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

Field A3 - 05/26/2023: Corn, silage

Field name: Field A3 Crop: Corn, silage Plant date: 05/26/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	8.00	0.00	0.00	0.00
Dry manure	263.63	73.95	236.49	3,809.08
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	873.02
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	278.63	73.95	236.49	4,682.09
Anticipated crop nutrient removal	208.00	39.00	171.60	0.00
Actual crop nutrient removal	211.68	39.20	251.44	1,058.40
Nutrient balance	66.95	34.75	-14.95	3,623.69
Applied to removed ratio	1.32	1.89	0.94	4.42

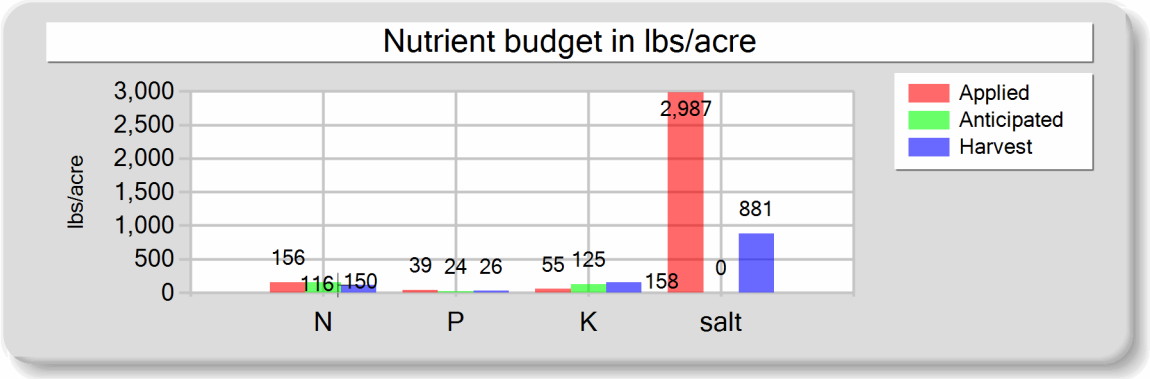
Fresh water applied	
40,800,000.00	*gallons*
1,502.53	*acre-inches*
38.53	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field A4 - 11/16/2022: Oats, silage-soft dough

Field name: Field A4 Crop: Oats, silage-soft dough Plant date: 11/16/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	148.72	39.24	54.87	2,986.97
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	155.72	39.24	54.87	2,986.97
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	115.86	25.62	157.78	880.79
Nutrient balance	39.86	13.63	-102.92	2,106.18
Applied to removed ratio	1.34	1.53	0.35	3.39

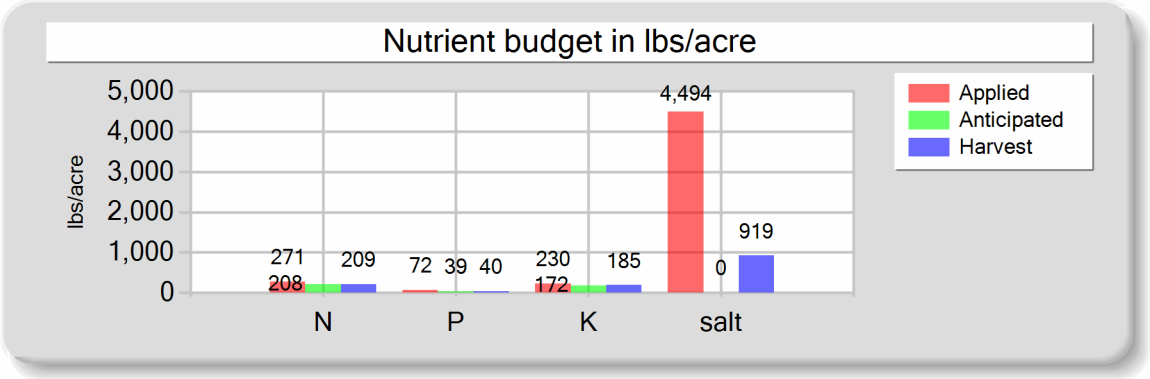
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

Field A4 - 05/25/2023: Corn, silage

Field name: Field A4 Crop: Corn, silage Plant date: 05/25/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	8.00	0.00	0.00	0.00
Dry manure	255.90	71.78	229.56	3,697.34
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	796.67
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	270.90	71.78	229.56	4,494.01
Anticipated crop nutrient removal	208.00	39.00	171.60	0.00
Actual crop nutrient removal	208.88	39.76	185.36	919.30
Nutrient balance	62.02	32.02	44.20	3,574.72
Applied to removed ratio	1.30	1.81	1.24	4.89

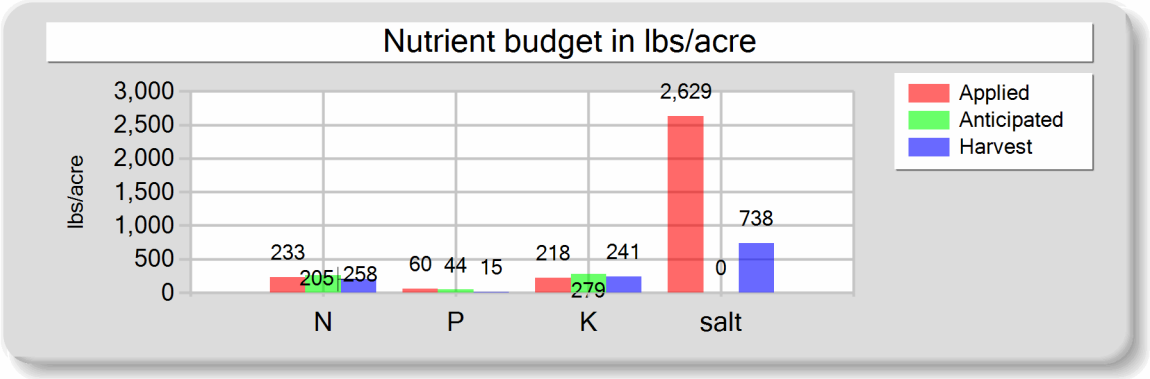
Fresh water applied	
42,960,000.00	*gallons*
1,582.07	*acre-inches*
35.16	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field B1 - 11/10/2016: Almond, in shell

Field name: Field B1 Crop: Almond, in shell Plant date: 11/10/2016



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	64.00	0.00	0.00	0.00
Dry manure	154.66	59.61	218.21	1,524.10
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	1,104.67
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	232.66	59.61	218.21	2,628.77
Anticipated crop nutrient removal	258.00	43.80	279.00	0.00
Actual crop nutrient removal	205.32	14.50	241.28	737.54
Nutrient balance	27.34	45.11	-23.07	1,891.23
Applied to removed ratio	1.13	4.11	0.90	3.56

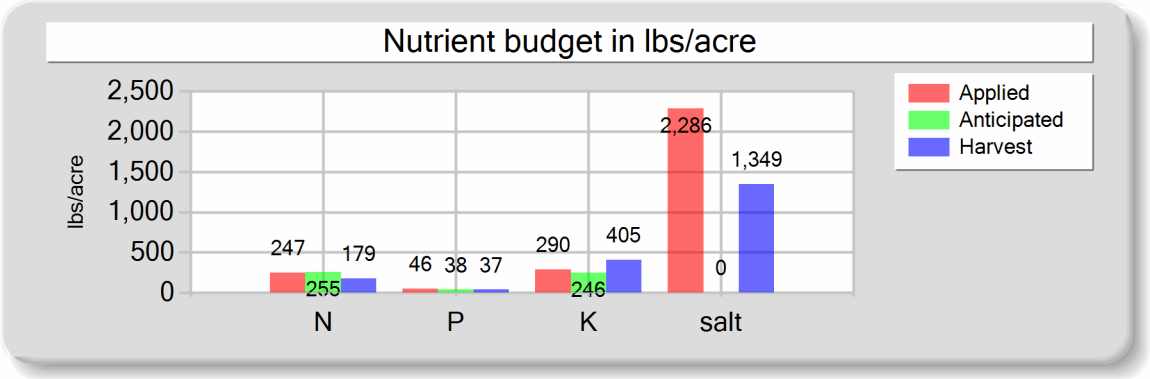
Fresh water applied	
19,856,230.00	*gallons*
731.24	*acre-inches*
48.75	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field B2 - 11/14/2022: Rye grass

Field name: Field B2 Crop: Rye grass Plant date: 11/14/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	239.90	46.34	289.70	2,286.44
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	246.90	46.34	289.70	2,286.44
Anticipated crop nutrient removal	255.00	37.50	246.00	0.00
Actual crop nutrient removal	179.18	37.40	404.60	1,349.46
Nutrient balance	67.72	8.94	-114.90	936.98
Applied to removed ratio	1.38	1.24	0.72	1.69

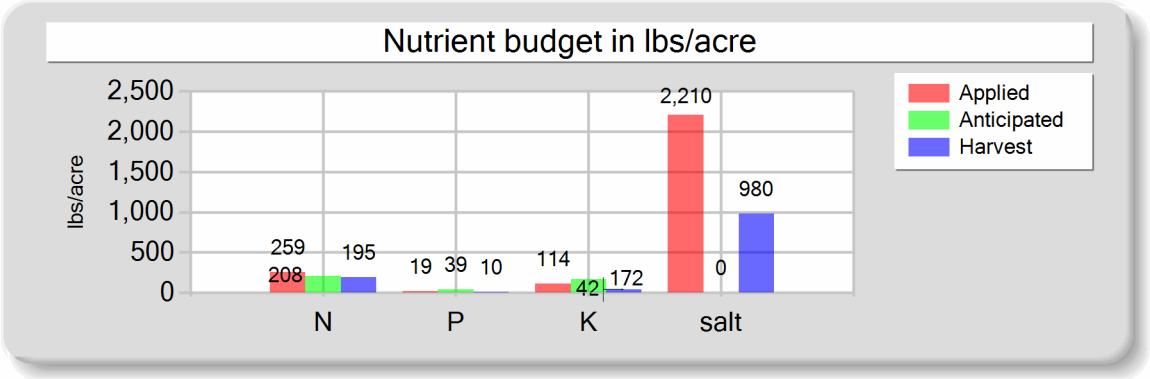
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
1,920,000.00	*gallons*
70.71	*acre-inches*
1.91	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field B2 - 06/07/2023: Corn, silage

Field name: Field B2 Crop: Corn, silage Plant date: 06/07/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	8.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	243.58	18.94	113.67	1,254.46
Fresh water	0.00	0.00	0.00	955.39
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	258.58	18.94	113.67	2,209.85
Anticipated crop nutrient removal	208.00	39.00	171.60	0.00
Actual crop nutrient removal	194.66	10.00	42.34	979.79
Nutrient balance	63.92	8.94	71.33	1,230.06
Applied to removed ratio	1.33	1.89	2.68	2.26

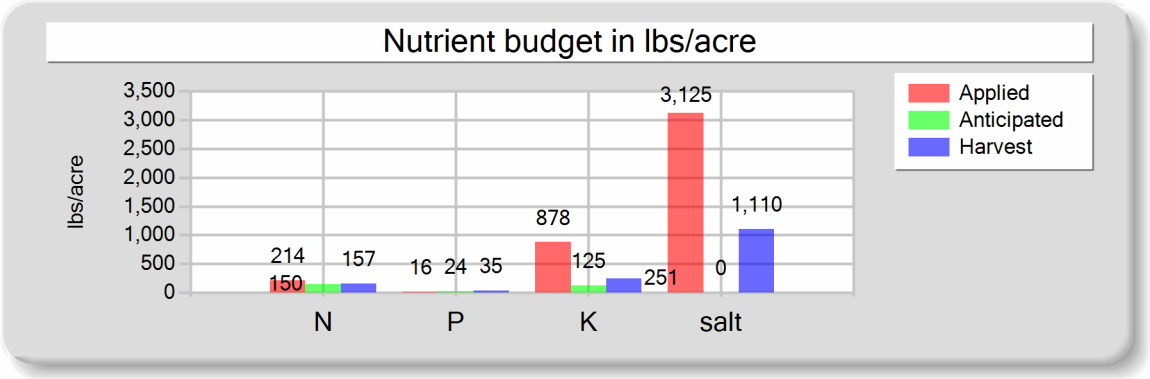
Fresh water applied	
42,360,000.00	*gallons*
1,559.97	*acre-inches*
42.16	*inches/acre*

Process wastewater applied	
900,000.00	*gallons*
33.14	*acre-inches*
0.90	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field B3 - 11/15/2022: Oats, silage-soft dough

Field name: Field B3 Crop: Oats, silage-soft dough Plant date: 11/15/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	207.29	16.50	877.71	3,124.91
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	214.29	16.50	877.71	3,124.91
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	156.65	34.81	250.64	1,110.19
Nutrient balance	57.64	-18.31	627.08	2,014.72
Applied to removed ratio	1.37	0.47	3.50	2.81

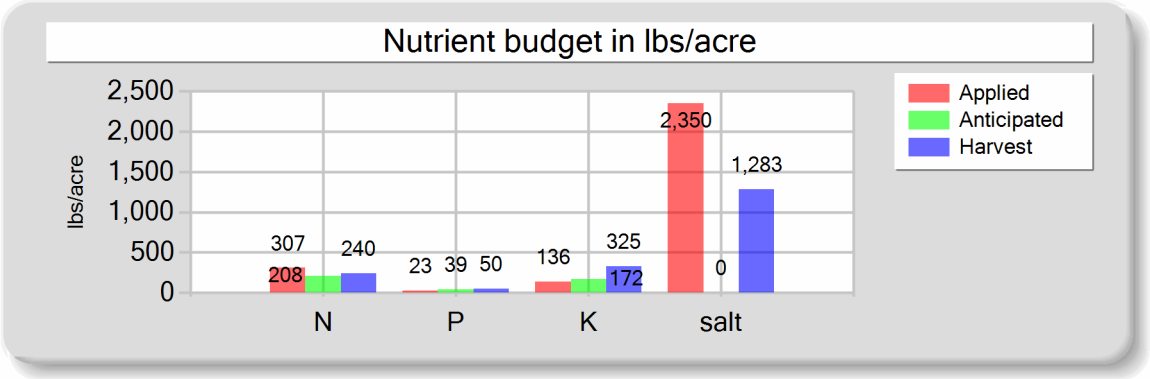
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
5,520,000.00 *gallons*
203.28 *acre-inches*
2.75 *inches/acre*

Total harvests for the crop
1 *harvests*

Field B3 - 05/24/2023: Corn, silage

Field name: Field B3 Crop: Corn, silage Plant date: 05/24/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	8.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	292.30	22.73	136.41	1,505.35
Fresh water	0.00	0.00	0.00	844.42
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	307.30	22.73	136.41	2,349.77
Anticipated crop nutrient removal	208.00	39.00	171.60	0.00
Actual crop nutrient removal	240.00	49.80	325.20	1,283.40
Nutrient balance	67.30	-27.07	-188.79	1,066.37
Applied to removed ratio	1.28	0.46	0.42	1.83

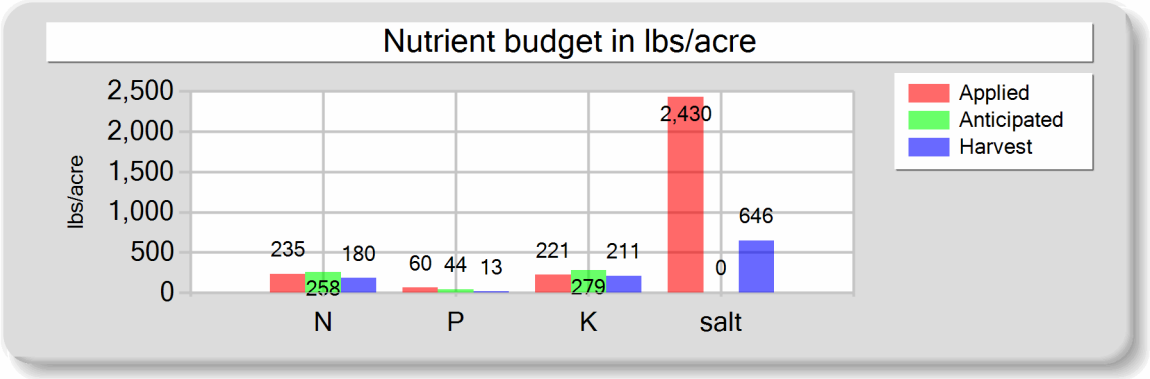
Fresh water applied	
74,880,000.00	*gallons*
2,757.58	*acre-inches*
37.26	*inches/acre*

Process wastewater applied	
2,160,000.00	*gallons*
79.55	*acre-inches*
1.07	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field D1 - 11/10/2016: Almond, in shell

Field name: Field D1 Crop: Almond, in shell Plant date: 11/10/2016



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	64.00	0.00	0.00	0.00
Dry manure	156.75	60.42	221.16	1,544.70
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	885.61
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	234.75	60.42	221.16	2,430.31
Anticipated crop nutrient removal	258.00	43.80	279.00	0.00
Actual crop nutrient removal	179.87	12.70	211.37	646.12
Nutrient balance	54.88	47.72	9.79	1,784.19
Applied to removed ratio	1.31	4.76	1.05	3.76

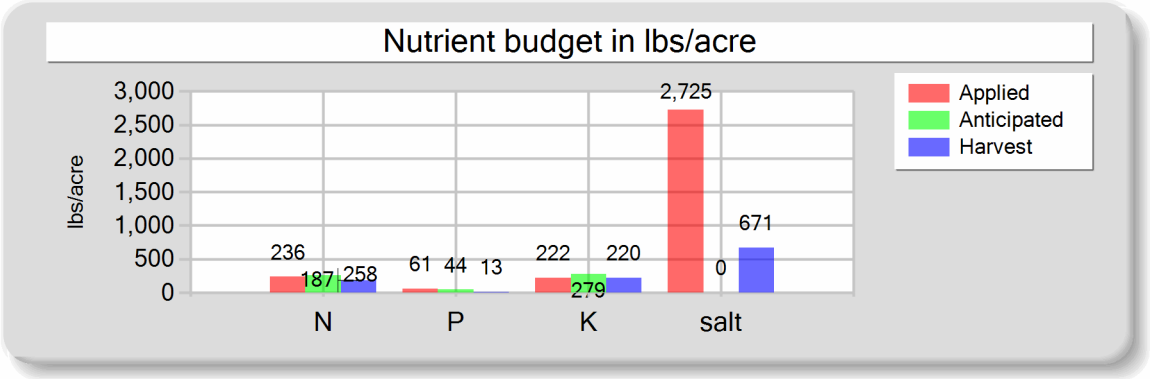
Fresh water applied	
39,266,260.00	*gallons*
1,446.04	*acre-inches*
39.08	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field D2 - 10/30/2018: Almond, in shell

Field name: Field D2 Crop: Almond, in shell Plant date: 10/30/2018



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	64.00	0.00	0.00	0.00
Dry manure	157.52	60.72	222.25	1,552.33
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	1,172.56
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	235.52	60.72	222.25	2,724.89
Anticipated crop nutrient removal	258.00	43.80	279.00	0.00
Actual crop nutrient removal	186.83	13.19	219.56	671.13
Nutrient balance	48.69	47.52	2.70	2,053.76
Applied to removed ratio	1.26	4.60	1.01	4.06

Fresh water applied	
75,875,592.00	*gallons*
2,794.24	*acre-inches*
51.75	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Corral Solids

Sample and source description: Corral Solids

Sample date: 10/13/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 6.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	24,700.00	7,460.00	26,000.00							55.60
DL	100.00	100.00	100.00							0.01

Corral Solids

Sample and source description: Corral Solids

Sample date: 10/14/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 29.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	27,500.00	10,600.00	38,800.00							27.10
DL	100.00	100.00	100.00							0.01

Settling Basin

Sample and source description: Settling Basin

Sample date: 10/14/2022 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 63.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	23,800.00	6,280.00	8,780.00							47.80
DL	100.00	100.00	100.00							0.01

Corral Manure

Sample and source description: Corral Manure

Sample date: 04/28/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 24.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	27,200.00	7,630.00	24,400.00							39.30
DL	100.00	100.00	100.00							0.01

B. PROCESS WASTEWATER ANALYSES

WWQ4

Sample and source description: WWQ4

Sample date: 10/21/2022 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	333.00	146.00	0.00	0.00	26.50	1,410.00								8,140.00	5,020
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWQ1

Sample and source description: WWQ1

Sample date: 01/20/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	554.00	235.00	0.00	0.00	107.00	669.00								5,840.00	5,280
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWQ2

Sample and source description: WWQ2

Sample date: 05/19/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1,200.00	398.00	0.00	0.00	93.30	560.00	151.00	168.00	222.00	4,800.00	0.00	268.00	439.00	6,160.00	6,180
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

WWQ3

Sample and source description: WWQ3

Sample date: 07/27/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	342.00	185.00	0.00	0.00	21.50	915.00								6,860.00	6,680
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWQ4

Sample and source description: WWQ4

Sample date: 10/27/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	538.00	361.00	0.00	0.00	61.20	578.00								6,460.00	5,280
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

C. FRESH WATER ANALYSES

River

River

River

Sample description: River

Sample date: 07/27/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								163.00	100
DL	1.00	0.50	0.50								10.00	10

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

Field A1 - 11/16/2022: Oats, hay

Forage

Sample and source description: Forage

Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 16.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	24,000.00	2,160.00	19,900.00		8.86
DL	100.00	100.00	100.00		0.01

Field A1 - 06/07/2023: Corn, silage

Field A1 - 06/07/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 11/10/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 72.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,340.00	810.00	3,340.00		6.40
DL	100.00	100.00	100.00		0.01

Field A2 - 11/16/2022: Oats, hay

Forage

Sample and source description: Forage

Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 8.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,000.00	2,750.00	25,900.00		9.60
DL	100.00	100.00	100.00		0.01

Field A2 - 06/07/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 11/10/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 72.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,340.00	810.00	3,340.00		6.40
DL	100.00	100.00	100.00		0.01

Field A3 - 11/16/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/05/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 69.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,490.00	1,040.00	7,050.00		12.10
DL	100.00	100.00	100.00		0.01

Field A3 - 05/26/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/29/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,780.00	700.00	4,490.00		6.30
DL	100.00	100.00	100.00		0.01

Field A4 - 11/16/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/05/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 75.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,980.00	880.00	5,420.00		12.40
DL	100.00	100.00	100.00		0.01

Field A4 - 05/25/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/29/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 65.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,730.00	710.00	3,310.00		4.80
DL	100.00	100.00	100.00		0.01

Field B1 - 11/10/2016: Almond, in shell

Almonds

Sample and source description: Almonds

Sample date: 09/06/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 11.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,700.00	1,250.00	20,800.00		7.16
DL	100.00	100.00	100.00		0.01

Field B2 - 11/14/2022: Rye grass

Forage

Sample and source description: Forage

Sample date: 05/05/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,270.00	1,100.00	11,900.00		13.50
DL	100.00	100.00	100.00		0.01

Field B2 - 06/07/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 11/10/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 76.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,310.00	170.00	720.00		7.00
DL	100.00	100.00	100.00		0.01

Field B3 - 05/24/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 05/05/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 73.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,480.00	960.00	7,240.00		11.90
DL	100.00	100.00	100.00		0.01

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/29/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 69.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,000.00	830.00	5,420.00		6.90
DL	100.00	100.00	100.00		0.01

Field D1 - 11/10/2016: Almond, in shell

Field D1 - 11/10/2016: Almond, in shell

Almonds

Sample and source description: Almonds

Sample date: 09/06/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 11.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,700.00	1,250.00	20,800.00		7.16
DL	100.00	100.00	100.00		0.01

Field D2 - 10/30/2018: Almond, in shell

Almonds

Sample and source description: Almonds

Sample date: 09/06/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 11.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,700.00	1,250.00	20,800.00		7.16
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

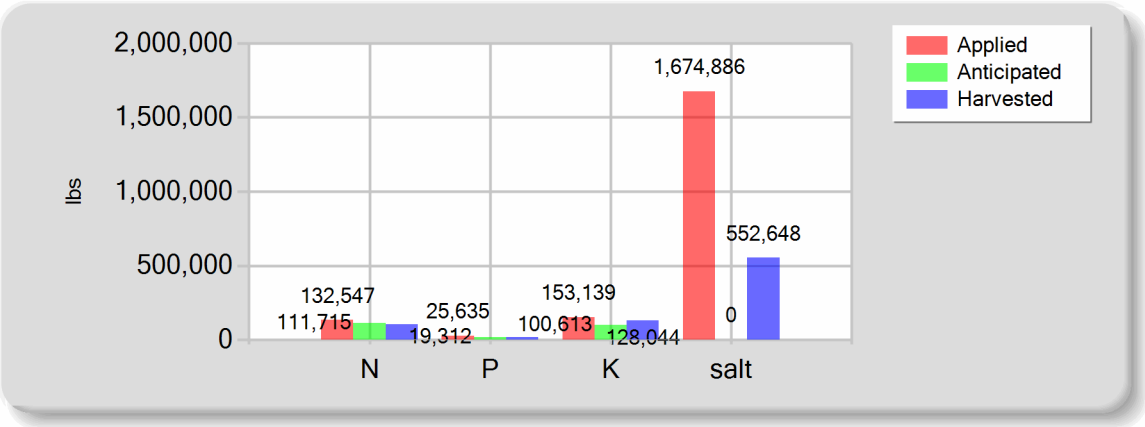
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

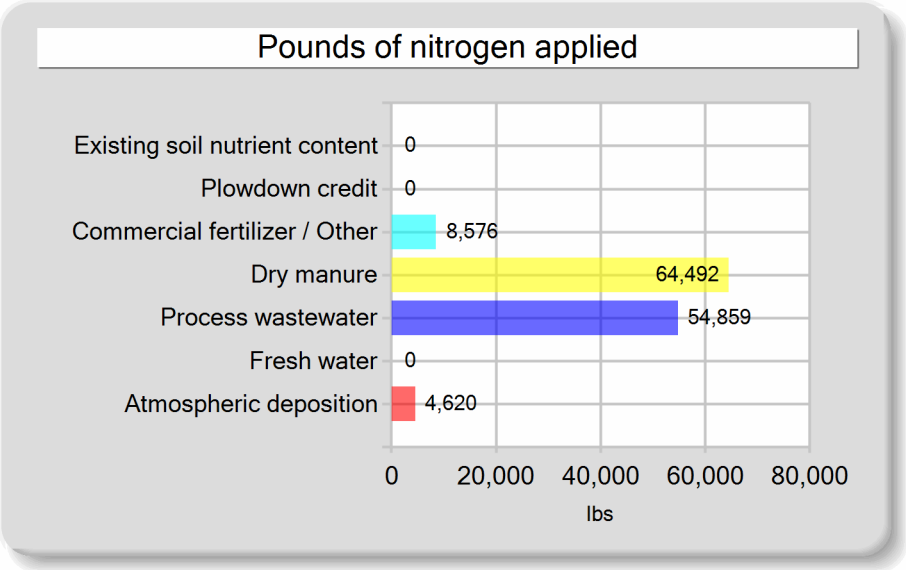
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

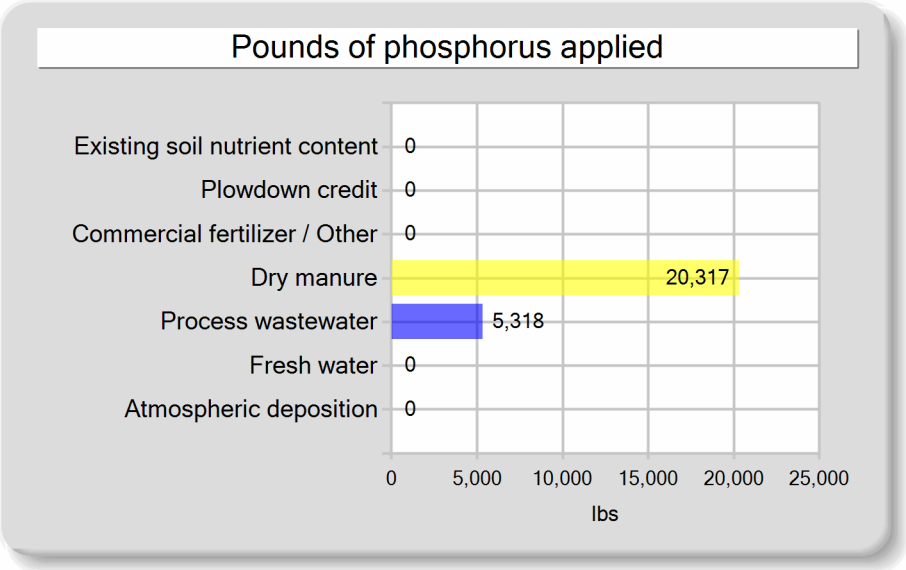
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	8,576.00	0.00	0.00	0.00
Dry manure	64,492.00	20,317.26	63,169.65	895,708.00
Process wastewater	54,858.69	5,317.58	89,969.78	473,652.19
Fresh water	0.00	0.00	0.00	305,525.54
Atmospheric deposition	4,620.00	0.00	0.00	0.00
Total nutrients applied	132,546.69	25,634.85	153,139.43	1,674,885.73
Anticipated crop nutrient removal	111,715.00	19,312.30	100,613.40	0.00
Actual crop nutrient removal	102,569.72	17,164.76	128,044.06	552,648.20
Nutrient balance	29,976.97	8,470.09	25,095.37	1,122,237.52
Applied to removed ratio	1.29	1.49	1.20	3.03

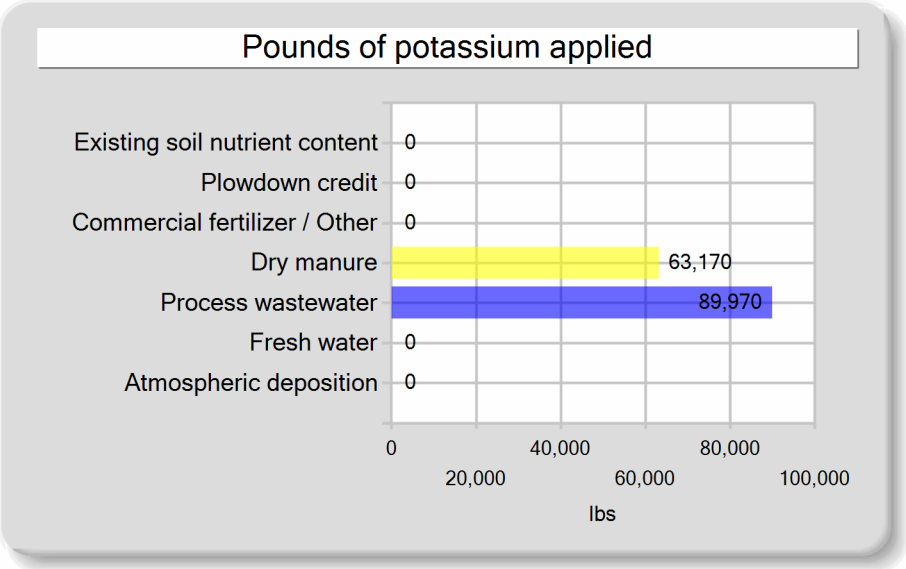
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

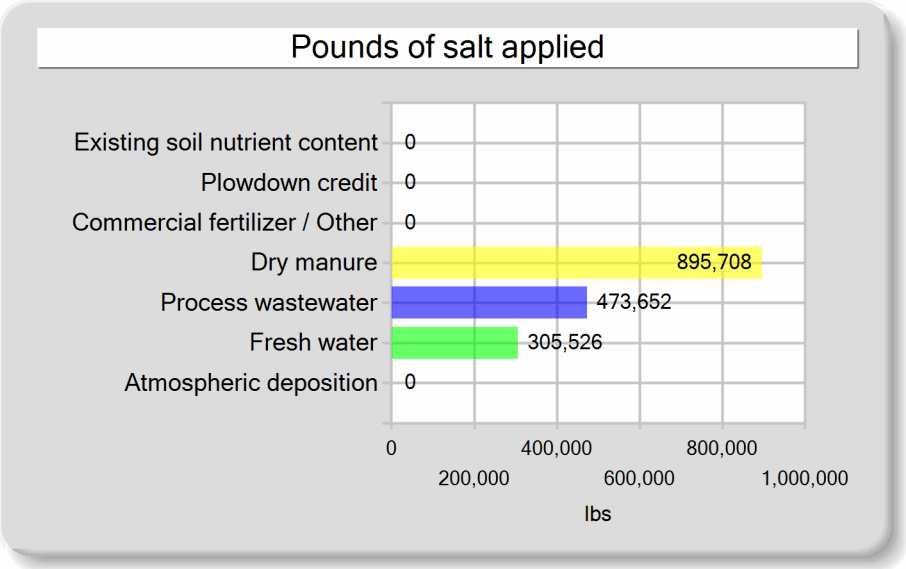


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Arlan Vander Woude
Arlan Vander Woude (Jun 14, 2024 10:01 PDT)

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Arlan Vander Woude	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
Jun 14, 2024	
DATE	DATE

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

www.denelelabs.com

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
Delta Woods
18700 S. Bacchetti Rd.
Tracy, CA 95304

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 3/20/23 7:30 AM
SUBMITTED BY: Sean
ANALYZED DATE: 3/20/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW-Heifers	T3079001A	Electrical Conductivity (EC)	EPA 120.1 and	0.963	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	0.510	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	616	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	104	93.2 (T3079001A)	93.3 (T3079001A)	0.064
Electrical Conductivity (EC)	96.4			0.000
Nitrate Nitrogen (NO3-N)	104	102 (T3076025A)	101 (T3076025A)	1.52

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

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Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 3/20/23 7:30 AM
SUBMITTED BY: Sean
ANALYZED DATE: 3/20/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW-Milk Barn	T3079001B	Electrical Conductivity (EC)	EPA 120.1 and	1.02	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	1.89	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	652	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	104	93.2 (T3079001A)	93.3 (T3079001A)	0.064
Electrical Conductivity (EC)	96.4			0.000
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ELAP Certificate No.: 2714

DENELE ANALYTICAL, INC.
865 S. KILROY ROAD
TURLOCK, CA 95380
PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: DELTA WOODS
Mailing address:
(not facility address) TRACY CA
Phone#:

Authorized Copy Released To: F&R AG SERVICES
Sampled By: Sean Locke
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- H1 EC*, NO3-N, NH4-N*
- H6 EC, TN, TDS
- H7 Combination of H1 & H6
- H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- H3 EC*, NH4-N, TKN, TP, K, TDS
- H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- H8 Combination of H3 &H5
- H9 Combination of H4 &H5

TILE DRAIN
- H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
- E1 EC*, Temp*, pH*, NO3-N, NH3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- E2 EC,* Temp*, pH*, NO3-N, NH3-N*, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- E3 EC*, Temp*, pH*, NO3-N, NH3-N*, NH4-N*, TP, Total & Fecal Coliforms
- E4 EC*, Temp*, pH*, NO3-N, NH3-N*, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- M1 % Moisture
- M3 TN TP, K, % Moisture
- M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- M7 Combination of M3 & M4

PLANT TISSUE
- F18 % Moisture, Ash, TN, P, K, DM and as received
- F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- K3 Soluble Phosphorus
- K4 1ft NO3, Organic Matter
- K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
- K7 Combination of K3 & K4
- K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	Analysis: H1	H3	M3	F18	OTHER
1	3/17/23	8:55	DW - HEIFERS	DOMESTIC	3.0°C		x				
2	"	9:00	DW - MILK BARN	"	2.8°C		x				
3											
4											
5											
6											
7											
8											
9											
10											

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	(LABORATORY USE ONLY) Logged in By:	LABORATORY NUMBER
		F&R AG SERVICES				
2nd		DENELE ANALYTICAL, INC.	3-20-23 7:30			T3079001

Required to be submitted to any agency? (YES / NO)

2023 Delta Woods Annual Report

Final Audit Report 2024-06-14

Created:	2024-06-10
By:	Sean Locke (slocke@fragservices.com)
Status:	Signed
Transaction ID:	CBJCHBCAABAA0C7U29NbcQ7K3N1lricO-lzwBZg76aFF

"2023 Delta Woods Annual Report" History

Document created by Sean Locke (slocke@fragservices.com)
2024-06-10 - 10:50:28 PM GMT

Document emailed to deltawoodsdairy@gmail.com for signature
2024-06-10 - 10:51:10 PM GMT

Email viewed by deltawoodsdairy@gmail.com
2024-06-14 - 12:52:53 PM GMT

Signer deltawoodsdairy@gmail.com entered name at signing as Arlan Vander Woude
2024-06-14 - 5:01:22 PM GMT

Document e-signed by Arlan Vander Woude (deltawoodsdairy@gmail.com)
Signature Date: 2024-06-14 - 5:01:24 PM GMT - Time Source: server

Agreement completed.
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